

Quantum and AI Waltz

Solving Systems of Linear Equations Using Sample-Based Quantum Diagonalization

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Research/Lecturer of Quantum Computing and Quantum Theoretical Chemistry at Amsterdam
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15th August 2026
Quantum AI and NLP Conference, Aug 14-16, 2026
Indiana University Indianapolis

- Researcher in **quantum information** and **quantum machine learning**, at the intersection of quantum mechanics, quantum chemistry, and machine learning
- Ph.D., Radboud University; M.Sc., University of Burgundy; B.Sc., American University in Cairo
- Lecturer and Researcher at the **Amsterdam University of Applied Sciences** of Quantum Information, Algorithms, and Quantum Theoretical Chemistry
- General Manager of **MolKet SAS**, an AI-driven consultancy and software startup
- Expertise in quantum algorithms for chemistry and machine learning
- Invited and peer-reviewed speaker at international conferences
- Technology-focused YouTuber sharing content about quantum AI, tech, and science.

Contact Information

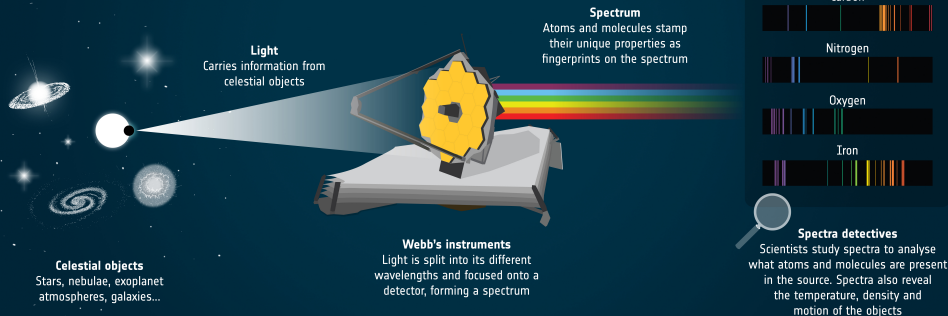
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- iQafé YouTube Channel: <https://www.youtube.com/@iQafe>

James Webb Space Telescope (JWST)

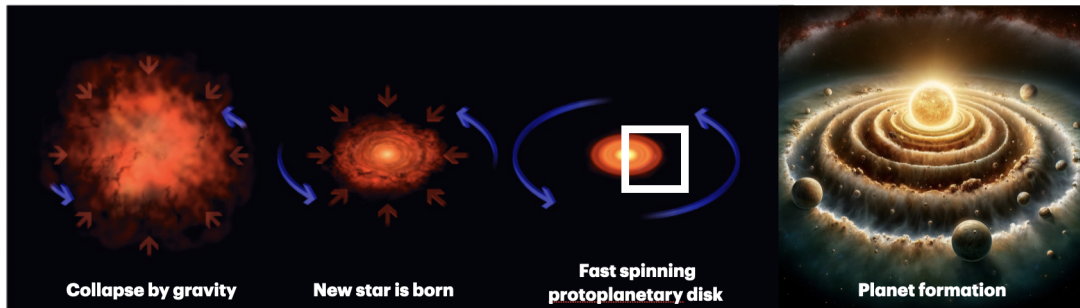


SPECTROSCOPY WITH WEBB

Spectroscopy is a tool that astronomers use to better understand the physics of objects in space. Like a prism splits white light from the Sun into its colour components (like a rainbow), Webb's spectrographs will dissect infrared light into its many wavelengths. This will provide detailed information about an object, such as how a galaxy moves or what molecules are present in an exoplanet's atmosphere.



Protoplanetary disks and molecular spectroscopy



© Addison-Wesley Longman

Conceptual illustration

Key molecules

CO CO₂ HCN H₂O C₂H₂ CH₄

Simulating CO₂ + He inelastic collisions

Second, we construct the 3D **Hamiltonian** of the dimer $\hat{H}_{HeCO_2}$ to obtain the **scattering wavefunctions** and **channel basis**:

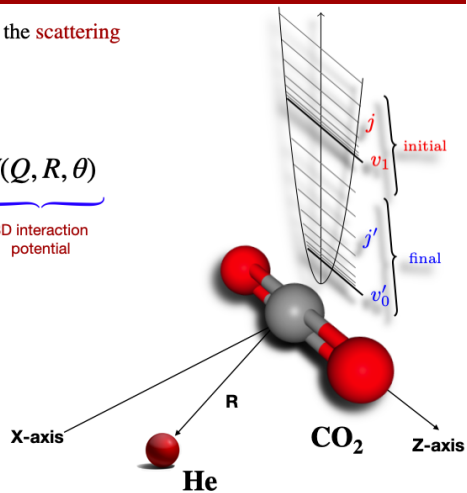
$$\hat{H}_{HeCO_2} = \underbrace{-\frac{1}{2\mu R} \frac{\partial^2}{\partial R^2} R}_{\text{Kinetic energy of the dimer}} + \underbrace{\hat{H}_{CO_2}}_{CO_2 \text{ Hamiltonian}} + \underbrace{\frac{\hat{L}^2}{2\mu R^2}}_{\text{Angular kinetic energy of the dimer}} + \underbrace{V(Q, R, \theta)}_{\text{3D interaction potential}}$$

dimer: refers to both He+CO₂

HeCO₂ computational basis/space, used for solution:

Space-fixed (SF): $|v \ j \ L \ |JM_J\rangle$

Body-fixed (BF): $|v \ j \ \Omega \ |JM_J\rangle$



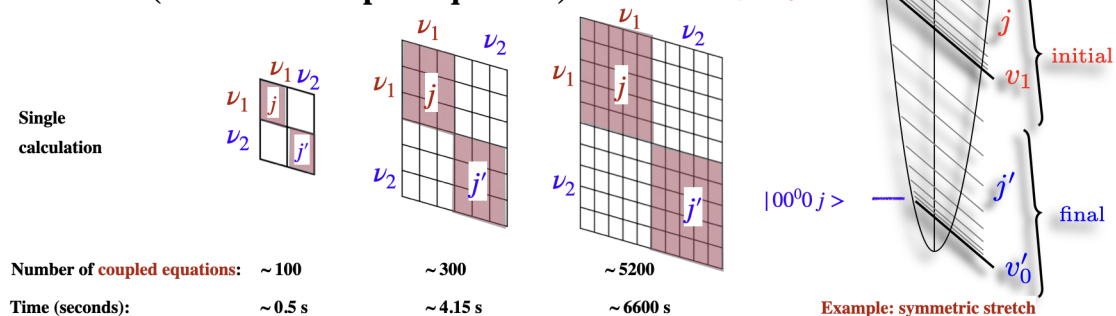
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Computational complexity of rovibrational of transition rates

Two main challenges, considering CO₂:

- Theoretical formalism and codes to enable the calculations
- Efficient procedures to reduce the computational complexity

$$\text{Time} \propto (\text{Number of coupled equations})^3$$

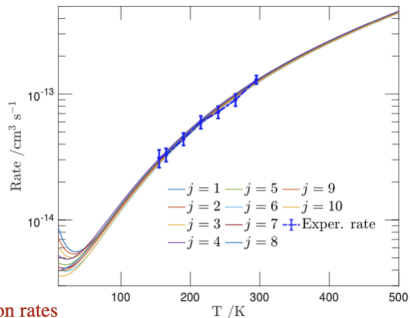


PhD work results

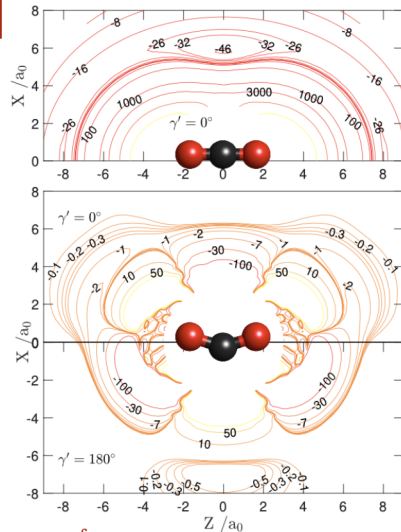
PhD work

Develop theoretical formalisms and codes to calculate rates of collision-induced rovibrational transitions of **CO₂** with **He** atoms.

Calculations: Symmetric, asymmetric, and bend modes



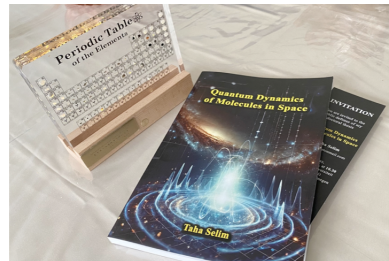
Example,
bend collision rates



Example,
bend potential energy surface

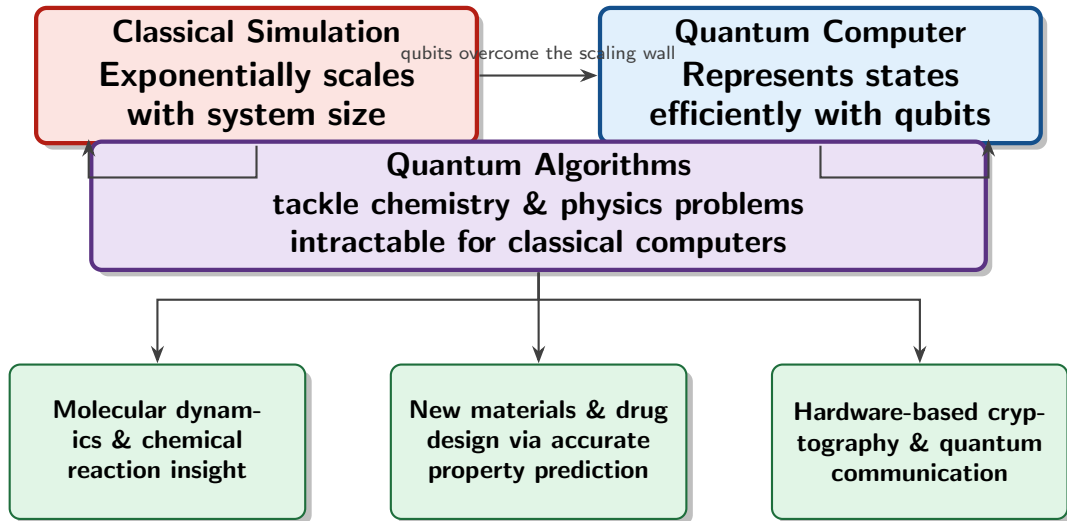
PhD Research Overview

- Taha Selim. (2024). *Quantum dynamics of molecules in space: Theoretical studies and efficient computational methods for collision-induced rovibrational transition rates in molecules*. [Doctoral dissertation, Radboud University].
- Taha Selim, Arthur Christianen, Ad van der Avoird, & Gerrit C. Groenenboom. (2021). Multi-channel distorted-wave Born approximation for rovibrational transition rates in molecular collisions. *The Journal of Chemical Physics*, 155(3), 034105.
- Taha Selim, Ad van der Avoird, & Gerrit C. Groenenboom. (2022). Efficient computational methods for rovibrational transition rates in molecular collisions. *The Journal of Chemical Physics*, 157(6), 064105.
- Taha Selim, Ad van der Avoird, & Gerrit C. Groenenboom. (2023). State-to-state rovibrational transition rates for C_3 in the bend mode in collisions with He atoms. *The Journal of Chemical Physics*, 159(16), 164310.



PhD Thesis Cover Page,
Taha Selim, 2024

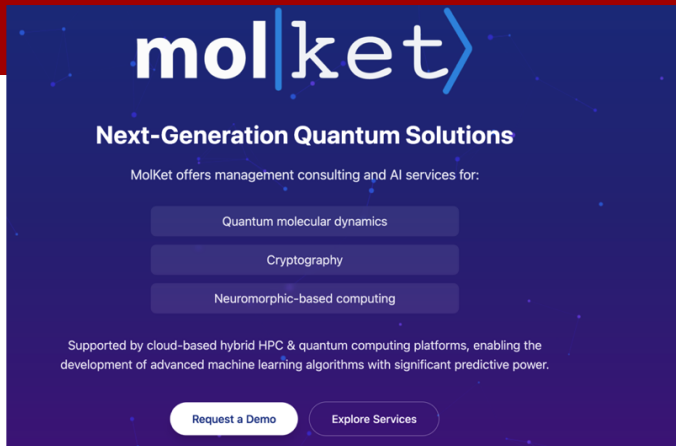
Why Quantum Simulations?



Computational complexity of molecular modelling and simulations

- Molecular systems are inherently quantum mechanical, requiring accurate quantum descriptions for properties and dynamics.
- Atomic transitions also cost extensive computational resources to model accurately.
- Multi-electron calculations scale exponentially with the number of electrons.
- Classical computers struggle with simulating quantum systems due to this exponential scaling.
- Quantum computers can potentially simulate quantum systems more efficiently by leveraging qubits and quantum algorithms

Can quantum computing and AI help us solve the computational complexity of molecular modelling and simulations?

The image shows a website banner for MolKet. The background is dark blue with a subtle pattern of white dots and lines, resembling a molecular structure or a network. At the top, the MolKet logo is displayed in white, with the 'k' and 'e' in a stylized font. Below the logo, the text 'Next-Generation Quantum Solutions' is written in white. Underneath, a line of text states 'MolKet offers management consulting and AI services for:'. This is followed by three rounded rectangular buttons stacked vertically, each containing one of the following services: 'Quantum molecular dynamics', 'Cryptography', and 'Neuromorphic-based computing'. Below these buttons, a paragraph of text reads: 'Supported by cloud-based hybrid HPC & quantum computing platforms, enabling the development of advanced machine learning algorithms with significant predictive power.' At the bottom of the banner, there are two rounded rectangular buttons: 'Request a Demo' and 'Explore Services'.

The company also offers consulting services for training AI on datasets from fields such as chemistry, biology, materials science, and cosmology. These services are supported by physics-based kernels (e.g., quantum-mechanical kernels), enabling the development of advanced machine learning algorithms with significant predictive power.



Dr. Taha Selim

Directeur Général – General Manager

CREDENTIALS

- Lecturer/Researcher (QC Education Officer), Amsterdam University of Applied Sciences
- MVP Founder Member QSECDEF
- PhD Quantum Theoretical & Computational Chemistry
- M.Sc. of Quantum physics, Lasers, & Materials

RECENT PUBLICATIONS

- Carbon dioxide (CO₂) as a quantum molecular sensor in protoplanetary disks, at APS March Meeting 2024
- State-to-state rovibrational transition rates for CO₂ in the bend mode in collisions with He atoms, J. Chem. Phys., 159, 164310 (2023)

RECENT TALKS

- Co-speaker lightning talk Simulating quantum molecular dynamics with Julia at JuliaCon 2024
- Speaker, Carbon dioxide (CO₂) as a quantum molecular sensor in protoplanetary disks, at APS March Meeting 2024
- Panelist at the QED-C Quantum & AI Use Cases Panel (2025)



Alain Chancé

Président – President

CREDENTIALS

- Principal consultant with 30+ years experience in major management consulting firms
- MVP Founder Member QSECDEF
- IEEE Senior Member
- Society affiliate American Chemical Society (ACS)
- Member American Physical Society (APS)
- Qiskit Advocate & IBM Certified Associate Developer
- Master's Degree in Science and Executive Engineering MINES Saint-Etienne, France (1981)

RECENT PUBLICATIONS

- QPP-Alain, repository of the conference paper Quantum Permutation Pad with Qiskit Runtime
- Quantum Permutation Pad Corinthians, In: Femmam, S., Lorenz, P. (eds) Recent Advances in Communication Networks and Embedded Systems
- Quantum Chemistry and Computing for the Curious: Illustrated with Python and Qiskit® code, Keeper L. Sharkey, Alain Chancé, Packt (2022)

RECENT TALKS

- Co-speaker lightning talk Simulating quantum



Example projects

- Develop quantum algorithms and software for chemistry- and physics-inspired use cases across various quantum platforms, and we provide benchmarking services
- Train Smart AI-assistants for chemistry and physics simulations and design
- Automate Hamiltonian constructions with AI



contact:

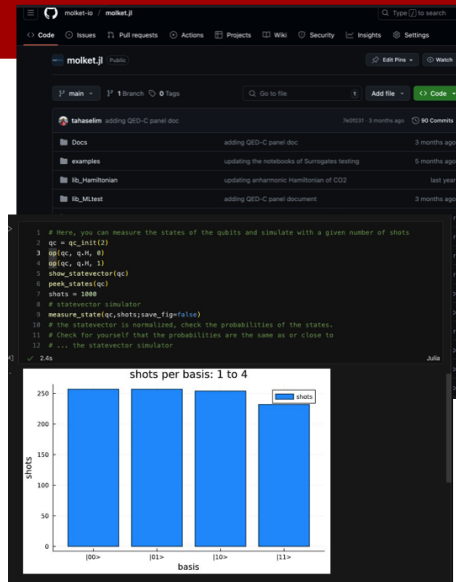
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MolKet's cloud:

www.molket.io

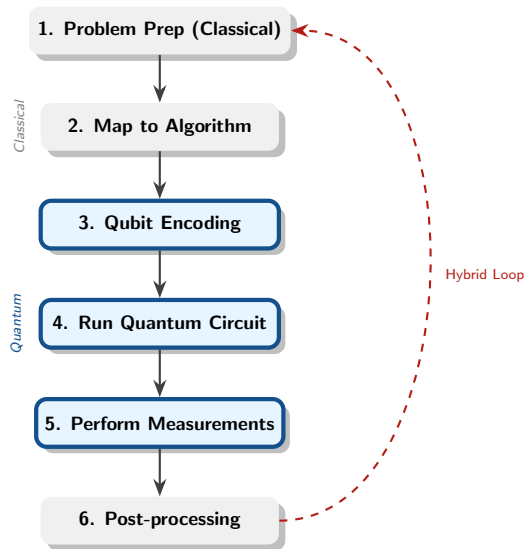
GitHub:

[molket-io/molket.jl](https://github.com/molket-io/molket.jl)

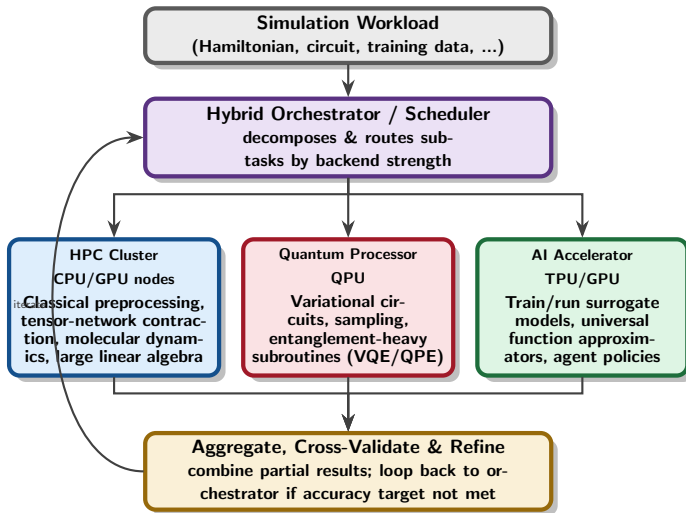


Quantum computing workflow diagram

- **Problem Preparation:**
Define Hamiltonian using classical algorithm.
- **Mapping:**
Convert problem to quantum compatible format (Hybrid workflow).
- **Data Loading:**
Initial state encoding on quantum register (Qubit encoding).
- **Execution:**
Translate algorithm to circuit and run on QPU.
- **Measurement:**
Collapse state and read classical bits.
- **Post-processing:**
Classical analysis of measurement results.



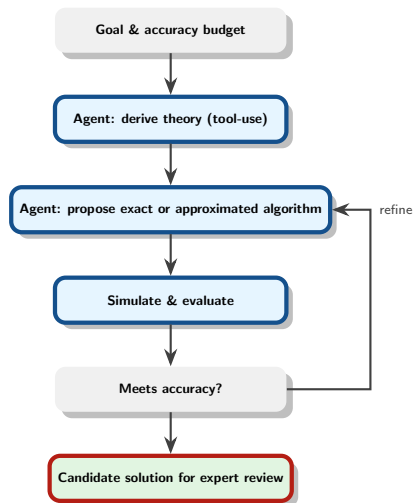
Distributed Execution: HPC, Quantum, and AI Accelerators



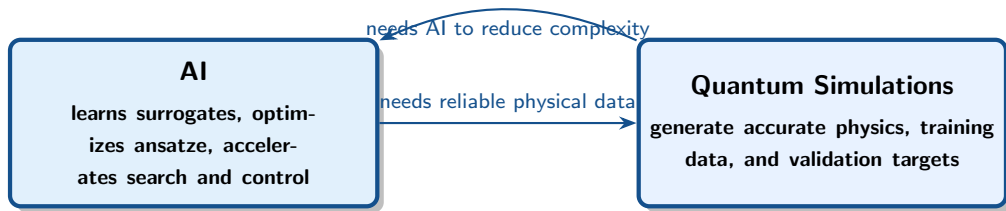
Agentic AI Loop for Quantum Simulation

Human-in-the-loop, not human-in-every-step:

- **Define goal:** target quantity (e.g. ground-state energy, transition rate) and accuracy budget.
- **Agent derives/retrieves theory:** symbolic tool-use to get the governing equations.
- **Agent proposes algorithm:** exact derivation *or* a universal-approximator shortcut.
- **Simulate & evaluate:** run on classical/quantum backend, score against validation checks.
- **Refine:** if accuracy is insufficient, agent revises the ansatz/approximation and loops.
- **Expert review:** human validates the final candidate before it is trusted.



AI and Quantum Simulations Need Each Other



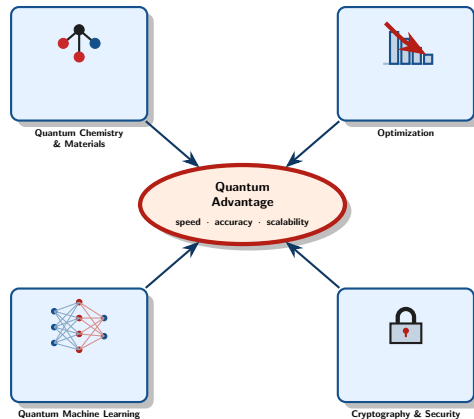
The Quest for Quantum Advantage

Noisy-intermediate-scale quantum (NISQ) era and beyond

In the NISQ era, quantum computers cannot yet outperform classical computers for most practical problems. Promising directions being explored include:

- **Quantum chemistry & materials:** simulating molecules, reactions, and materials.
- **Optimization:** logistics, finance, and machine-learning problems.
- **Quantum machine learning:** algorithms for data analysis, pattern recognition, generative modeling.
- **Cryptography & security:** quantum-resistant protocols and secure communication.

Quantum advantage: ongoing research aims for gains in speed, accuracy, and/or scalability.



Quantum Computational Chemistry and Quantum Simulations:

What is expected from quantum computing in the field of quantum chemistry and quantum simulations?

- Quantum computers: efficient representation of quantum states and operations.
- Reduce computational complexity of simulating quantum systems, especially for large molecules and materials.
- Example, from exponential to polynomial scaling in certain cases.
- Quantum chemistry algorithms can help in other fields like optimization, machine learning, and cryptography.
- Quantum advantage is problem-dependent and may not be achievable for all quantum chemistry problems.

Examples of potential quantum advantage in quantum chemistry include:

- Finding ground state energies of complex molecules.
- Potential Accurately estimating energy levels of quantum systems.
- Simulating chemical reactions and dynamics.
- Designing new materials and drugs with improved properties.
- Quantum control and optimization of quantum systems.

Hybrid Quantum-Classical Algorithms

For problems in quantum chemistry, quantum simulations, and optimization, hybrid quantum-classical algorithms are being developed. These algorithms leverage the strengths of both classical and quantum computing to achieve better performance.

Examples include:

- Variational Quantum Eigensolver (VQE): a hybrid algorithm for finding the ground state energy of a quantum system.
- Quantum Phase Estimation (QPE): a quantum algorithm for estimating the eigenvalues of a unitary operator.
- Quantum Approximate Optimization Algorithm (QAOA): a hybrid algorithm for solving combinatorial optimization problems.
- Quantum Machine Learning (QML): hybrid algorithms for training quantum neural networks and other machine learning models.

However, the computational complexity of such algorithms remains a challenge, and further research is needed to improve their efficiency and scalability.

Molecular Hamiltonian in brief

Hamiltonian, in general, is an operator that is defined as:

$$\hat{H} = \hat{T} + \hat{V}$$

where $\hat{T}$ is the kinetic energy operator and $\hat{V}$ is the potential energy operator.

The nuclear (vibrational) motion Hamiltonian for any polyatomic molecule is given by:

$$\hat{H}_{\text{nuc}} = \underbrace{-\frac{\hbar^2}{2} \sum_{i=1}^N \sum_{\alpha=1}^3 \frac{1}{M_i} \frac{\partial^2}{\partial R_{i\alpha}^2}}_{\text{Kinetic Energy}} + \underbrace{V(\mathbf{R}_1, \dots, \mathbf{R}_N)}_{\text{Potential Energy}}$$

where N is the number of nuclei, M_i is the mass of the i -th nucleus, $\mathbf{R}_i$ is the position of the i -th nucleus, and α is the Cartesian coordinate index. The vibrational (internal) part of the Hamiltonian for the nuclear motion

becomes in the harmonic approximation

$$\hat{H}_{\text{nuc}} \approx \frac{1}{2} \sum_{t=1}^{3N-6} \left[-\hbar^2 \frac{\partial^2}{\partial q_t^2} + f_t q_t^2 \right]$$

where q_t are the normal coordinates and f_t are the force constants.

Molecular Hamiltonian in brief

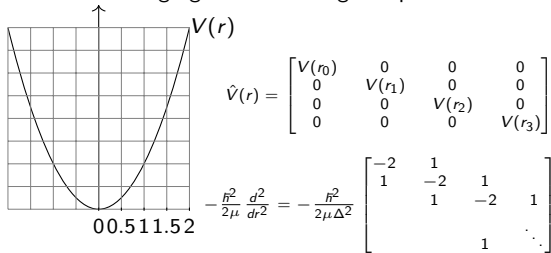
Typically, for 1D vibrations, we can represent the Hamiltonian as:

First quantized form: grid representation

$$\hat{H} = -\frac{\hbar^2}{2\mu} \frac{d^2}{dr^2} + \hat{V}(r)$$

This can be represented as a matrix based on a finite difference grid.

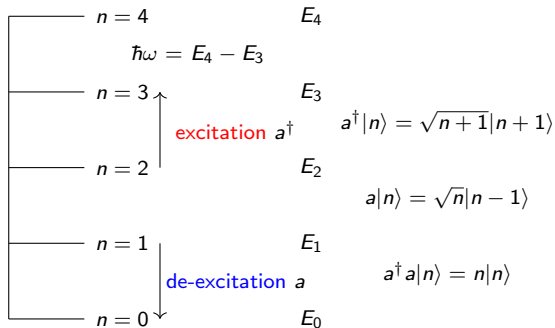
and the following figure shows the grid representation:



Second quantized form: basis set representation

$$\hat{H} = \hbar\omega(a^\dagger a + \frac{1}{2})$$

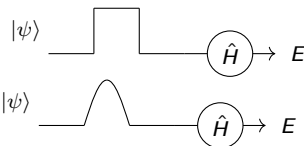
where $a^\dagger$ and a are the creation and annihilation operators, respectively.



Variational Principle

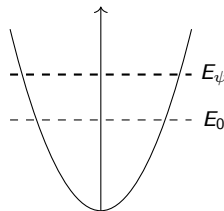
The action of the Hamiltonian on a wavefunction $|\psi\rangle$ yields the energy of the system.

- The eigenvalue equation: $\hat{H}|\psi\rangle = E|\psi\rangle$ yields the energy E of the system.
- The expectation value of the Hamiltonian is given by: $\langle\hat{H}\rangle = \langle\psi|\hat{H}|\psi\rangle = E_i$.



The **variational principle** states that the energy of any **trial wavefunction** is always greater than or equal to the true **ground state energy**.

- The trial wavefunction is denoted by $|\psi\rangle$.
- Variational principle: $\langle\psi|\hat{H}|\psi\rangle \geq E_0$, where E_0 is the ground state energy.
- The quantity $\langle\psi|\hat{H}|\psi\rangle = E_\psi$ where E_ψ is the upper bound to the ground state energy E_0 .
- Having a better trial wavefunction $|\psi\rangle$ yields a better approximation to the ground state energy.



Fermionic creation and annihilation operators

Two types of operators are used to represent the Hamiltonian of a quantum harmonic oscillator:

- Bosonic operators: $[a, a^\dagger] = 1$
- Fermionic operators: $[f, f^\dagger] = 1$

The Fermionic operators satisfy the anti-commutation relation.

Bosonic operators:

$$H = \hbar\omega(a^\dagger a + \frac{1}{2})$$

where $a^\dagger$ and a are the bosonic creation and annihilation operators, respectively.

$$[a, a^\dagger] = 1, \quad [a, a] = 0, \quad [a^\dagger, a^\dagger] = 0$$

Examples:

- molecular vibrations of diatomic molecules.
- phonons.

Fermionic operators:

$$H = \hbar\omega(f^\dagger f - \frac{1}{2})$$

where $f^\dagger$ and f are the fermionic creation and annihilation operators, respectively.

$$\{f, f^\dagger\} = 1 \quad \{f, f\} = 0 \quad \{f^\dagger, f^\dagger\} = 0$$

Examples:

- electrons in atoms and molecules.
- Majorana fermions.

Fermionic and Bosonic qubits

Traditional qubits

- Most quantum computers use qubits as two-level systems.
- They can be realized with physical systems such as electrons, but without exploiting fermionic or bosonic structure directly.

Fermionic qubits

- Leverage fermionic properties directly.
- Natural for quantum chemistry and materials simulation.
- Can make fermionic dynamics more native to the hardware model.
- Encoding fermionic information into standard qubits is often the difficult step.

Bosonic qubits

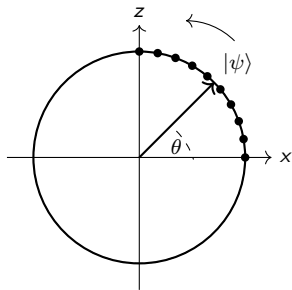
- Use bosonic systems directly, such as photons or harmonic-oscillator modes.
- Information lives in an infinite-dimensional Hilbert space.
- Offer promising routes for robust encoding, error correction, and fault tolerance.
- Let a bosonic state carry more structure than a simple two-level register.

Takeaway: the right qubit model should match the physics you want to simulate.

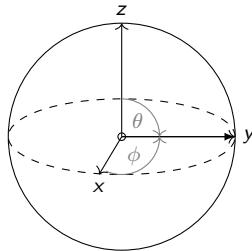
Variational calculations on a quantum computer

Variational quantum eigensolver (VQE): Makes use of parameterized quantum circuits to approximate the ground state energy of a given Hamiltonian.

Example: take $R_y(\theta)$ -gate as a parameterized gate to prepare the trial wavefunction.

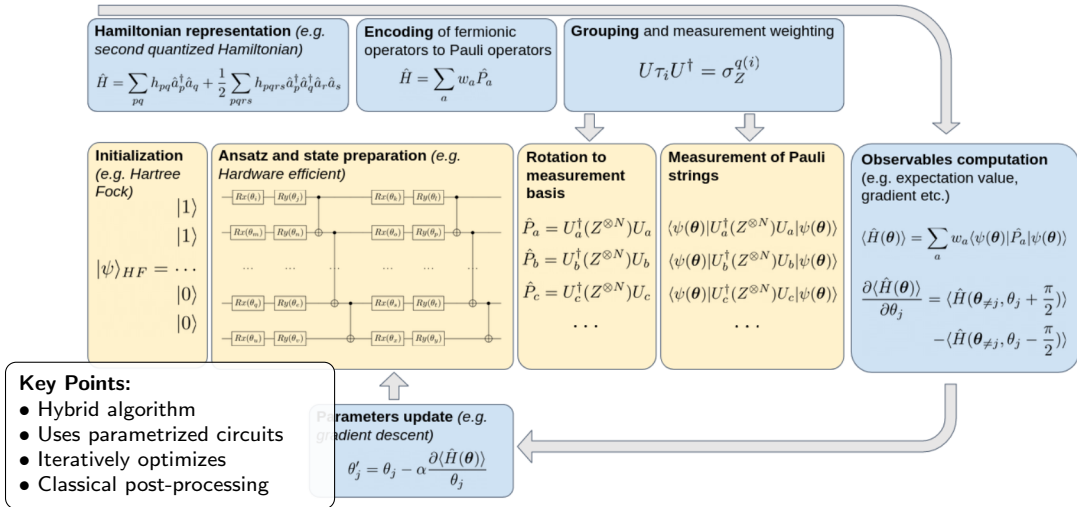


Aim: explore all possible states of a quantum system in order to find the ground state energy.



The parameter θ is optimized to minimize the energy of the system. The circuit spans n_{vqe} -state space.

Variational Quantum Eigensolver (VQE) hybrid workflow



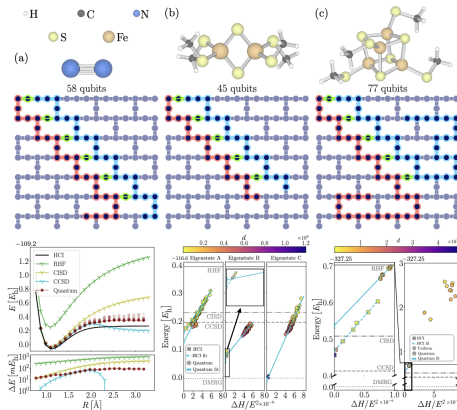
Computational complexity: parameters in the ansatz, variational iterations, and the number of measurements.

Use case I: IBM quantum-centric supercomputing architecture

IBM uses quantum-centric supercomputing architecture to tackle complex quantum chemistry problems: With HPC as Fugaku and QPU as Heron.

Recent IBM work using SQD to use chemistry on large basis set and compute:

- N_2 triple bond dissociation energy: well-known test of the accuracy of electronic structure methods in the presence of static electronic correlation.
- Ground states of Iron-Sulfur (FeS) clusters: The $[2\text{ Fe}-2\text{ S}]$ cluster and $[4\text{ Fe}-4\text{ S}]$ cluster, challenging systems for classical methods due to their complex electronic structure and strong electron correlation effects.

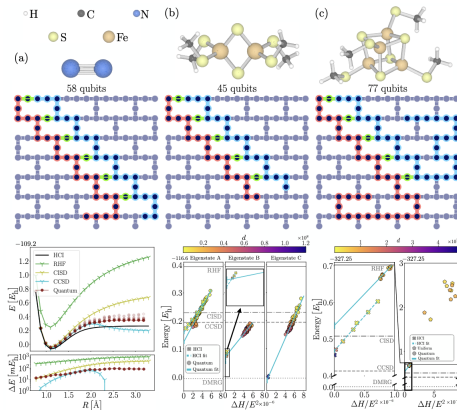


SQD use cases by IBM research team. Reference: [1]

Use case I: IBM quantum-centric supercomputing architecture

Nitrogen Molecule (N_2) Dissociation (6-31G basis)

Metric	Details
Qubits Used	36 total qubits used, with 32 for the Jordan-Wigner (JW) encoding.
Quantum Gates	762 CNOT gates. 1,408 total gates. Circuit depth $d = 148$.
HPC Usage (Classical)	Subspace diagonalization was performed on classical nodes using the Py-SCF library. A subspace dimension of $d = 4M$ was used for wavefunction visualization analysis.

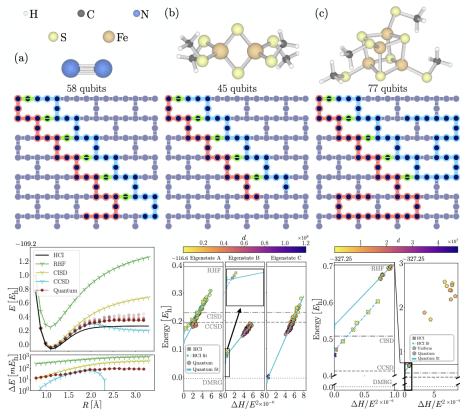


SQD use cases by IBM research team. [Reference: \[1\]](#)

Use case I: IBM quantum-centric supercomputing architecture

Nitrogen Molecule (N_2) Dissociation (cc-pVDZ basis)

Metric	Details
Qubits Used	58 total qubits used, with 52 for the Jordan-Wigner (JW) encoding.
Quantum Gates	1,792 CNOT gates. 3,412 total gates. Circuit depth $d = 223$. The circuit size reached approximately 1–1.5K two-qubit gates .
HPC Usage (Classical)	Used $d = 16 \cdot 10^6$ (16M) configurations for projection and diagonalization. The combined quantum runtime for all points in the dissociation curve was approximately 45 minutes . Classical diagonalization used the PySCF library on a single node.

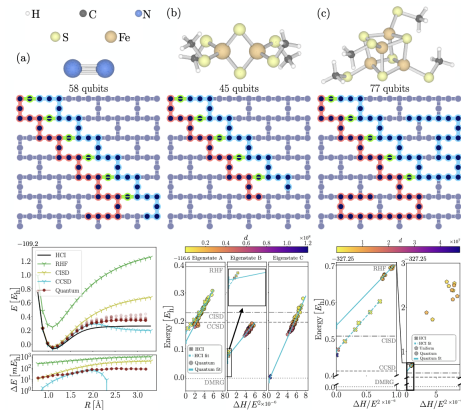


SQD use cases by IBM research team. Reference: [1]

Use case I: IBM quantum-centric supercomputing architecture

[2 Fe–2 S] Cluster Simulation

Metric	Details
Qubits Used	45 total qubits used, with 40 for the Jordan-Wigner (JW) encoding.
Quantum Gates	1,100 CNOT gates. 2,070 total gates. Circuit depth $d = 173$. Circuit sizes reached approximately 1–1.5K two-qubit gates .
HPC Usage (Classical)	Quantum runtime was approximately 45 minutes . Classical diagonalization used the PySCF library on a single node. Vertical scaling analysis showed the optimal configuration for this system required 30+ CPUs per node for the eigenstate solver.



SQD use cases by IBM research team. [Reference: \[1\]](#)

Why Solve $A\vec{x} = \vec{b}$? A Ubiquitous Problem

The problem: given a matrix $A \in \mathbb{C}^{N \times N}$ and vector $\vec{b} \in \mathbb{C}^N$, find $\vec{x}$ such that

$$A\vec{x} = \vec{b} \quad (\vec{x} = A^{-1}\vec{b} \text{ when } A \text{ is invertible}).$$

Simple to state, but N can be huge — and this exact problem sits at the core of three very different fields:

- **Quantum chemistry:** coupled-channel (close-coupling) equations in molecular collision theory — propagating and matching boundary conditions to extract scattering/reactance matrices requires solving large linear systems; coupled-perturbed Hartree–Fock/DFT response equations for molecular properties (polarizabilities, NMR shielding); discretized Poisson equation for electrostatics in DFT.
- **Quantum machine learning:** linear regression reduces to the normal equations $(X^\top X)\theta = X^\top Y$; kernel ridge regression solves $(K + \sigma^2 I)\alpha = \vec{y}$; second-order (Newton-type) training solves a Hessian linear system at every step.
- **Optimization:** Newton's method solves $H\Delta\vec{x} = -\nabla f$ each iteration; interior-point methods for LP/QP solve a KKT linear system per step; finite-element discretization of PDEs (structural, thermal, fluid) yields $K\vec{x} = \vec{f}$.

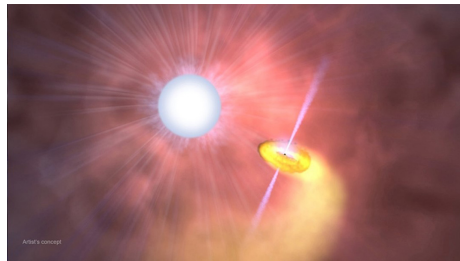
Common thread: N can reach millions (basis functions, features, mesh nodes). Classical direct solvers cost $O(N^3)$, iterative solvers $O(N\kappa)$ — quantum algorithms promise $O(\text{polylog } N)$ scaling under the right conditions (Sec. next).

X-ray astronomy simulations and black hole spin measurements

X-ray astronomy simulations help in studying underlying physics of the accretion of massive black holes, testing general relativity predictions in strong gravity fields, and measuring black hole spin rates through high-resolution spectroscopy.

- Investigating the extreme physics of black hole accretion to test predictions of General Relativity in strong gravity fields.
- Identifying the last stable orbit of matter around black holes, known as the Innermost Stable Circular Orbit (ISCO), before falling into the event horizon.
- The ISCO location directly correlates to spin. Rapidly spinning (*Kerr*) black holes pull the ISCO closer to the event horizon than static (*Schwarzschild*) ones.
- XRISM resolves fine spectral features (like Iron lines) distorted by extreme gravity and velocity, allowing precise calculation of spin rates for black holes of all sizes.

Special thanks to Jelle de Plaa (SRON), Liyi Gu (SRON), Dr. Marten Teitsma (HvA), Marc Bremer (HvA), Lourens Benningshof (BioDAC), and Jan Blommaart (BioDAC).



Artist impression of Cygnus X-3. The image illustrates a black hole candidate traversing the dense stellar winds of a Wolf-Rayet companion star. This environment is believed to sustain an accretion disk and launch powerful jets from the black hole. **Photo: NASA**

How does HHL work? — Problem Setup & Registers

HHL (Harrow, Hassidim, Lloyd) algorithm:

Given A , $\vec{b}$, solve $A\vec{x} = \vec{b}$. Classically $\vec{x} = A^{-1}\vec{b}$: Gaussian elimination is $O(N^3)$; with sparsity s and condition number $\kappa = \lambda_{\max}/\lambda_{\min}$, conjugate gradient reaches $O(Ns\kappa \log(1/\epsilon))$. HHL targets $O(\log N s^2 \kappa^2 \log(1/\epsilon))$.

Hermitian dilation: if A is not Hermitian, define

$$\tilde{A} = \begin{pmatrix} 0 & A^\dagger \\ A & 0 \end{pmatrix}, \quad \tilde{A} = \sum_i \lambda_i |u_i\rangle \langle u_i| \text{ (spectral decomposition).}$$

Expand $|b\rangle = \sum_i \beta_i |u_i\rangle$, so formally $|x\rangle = \tilde{A}^{-1}|b\rangle = \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$ — HHL prepares this *without* computing $\tilde{A}^{-1}$ explicitly.

Three registers:

- ANC: one ancilla qubit, $|0\rangle_{\text{ANC}}$.
- W: “work” register to hold QPE eigenvalue estimates, $|0\rangle_W$ (width set by desired precision).
- input–output register: holds the normalized $|\tilde{b}\rangle = \frac{|b\rangle}{\sqrt{\langle b|b\rangle}} = \sum_j \tilde{b}_j |u_j\rangle$, with $\tilde{b}_j = b_j / \sqrt{\langle b|b\rangle}$.

Initial state:

$$|\psi\rangle_0 = |0\rangle_{\text{ANC}} \otimes |0\rangle_W \otimes |\tilde{b}\rangle = |0\rangle_{\text{ANC}} \otimes \sum_j \tilde{b}_j |0\rangle_W \otimes |u_j\rangle$$

How does HHL work? — QPE, Rotation & Measurement

1. **Quantum phase estimation** with $U = e^{-i\tilde{A}t}$: since $e^{-i\tilde{A}t} = \sum_j e^{-i\lambda_j t} |u_j\rangle \langle u_j|$ and $e^{-i\lambda_j t} = e^{-2\pi i(\lambda_j t/2\pi)}$, QPE kicks back the phase $\tilde{\lambda}_j = \lambda_j t/2\pi$ into the work register:

$$|\psi\rangle_1 = |0\rangle_{\text{ANC}} \otimes \sum_j \tilde{b}_j |\tilde{\lambda}_j\rangle_W \otimes |u_j\rangle$$

2. **Conditional rotation**: rotate the ancilla about y , controlled on $|\tilde{\lambda}_j\rangle_W$, by $\theta_j = 2 \sin^{-1}(C/\tilde{\lambda}_j)$ — this is where $1/\lambda_j$ enters.

3. **Uncompute** (QPE^{-1}): the work register returns to $|0\rangle_W$ and decouples, leaving

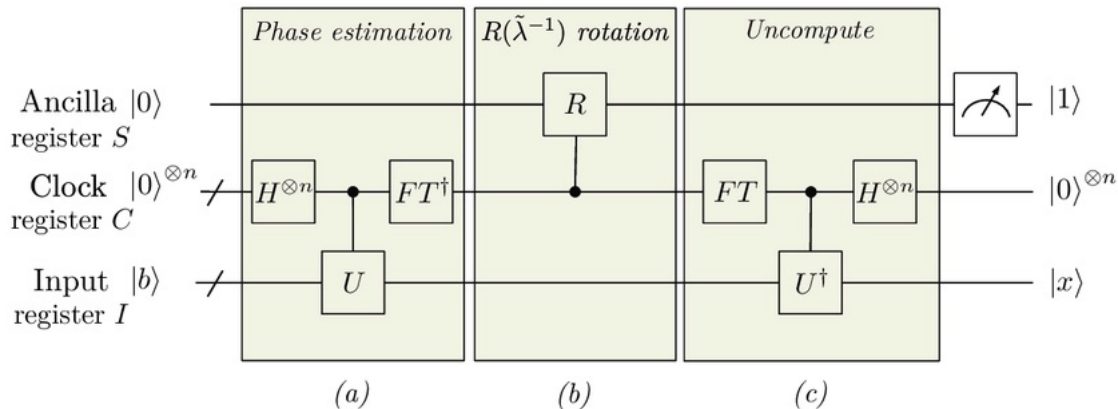
$$|\psi\rangle_5 = \sum_j \left(\tilde{b}_j \sqrt{1 - \frac{C^2}{\tilde{\lambda}_j^2}} |0\rangle_{\text{ANC}} \otimes |u_j\rangle + C \frac{\tilde{b}_j}{\tilde{\lambda}_j} |1\rangle_{\text{ANC}} \otimes |u_j\rangle \right)$$

4. **Measure the ancilla**; post-select $|1\rangle_{\text{ANC}}$:

$$|\psi\rangle_6 = C \sum_j \frac{\tilde{b}_j}{\tilde{\lambda}_j} |u_j\rangle = C \times \frac{2\pi}{t\sqrt{\langle b|b\rangle}} \sum_j \frac{b_j}{\lambda_j} |u_j\rangle \propto |x\rangle$$

C, t chosen so $C \times \frac{2\pi}{t\sqrt{\langle b|b\rangle}} \rightarrow 1$; post-selecting $|1\rangle_{\text{ANC}}$ succeeds with probability $O(1/\kappa^2)$.

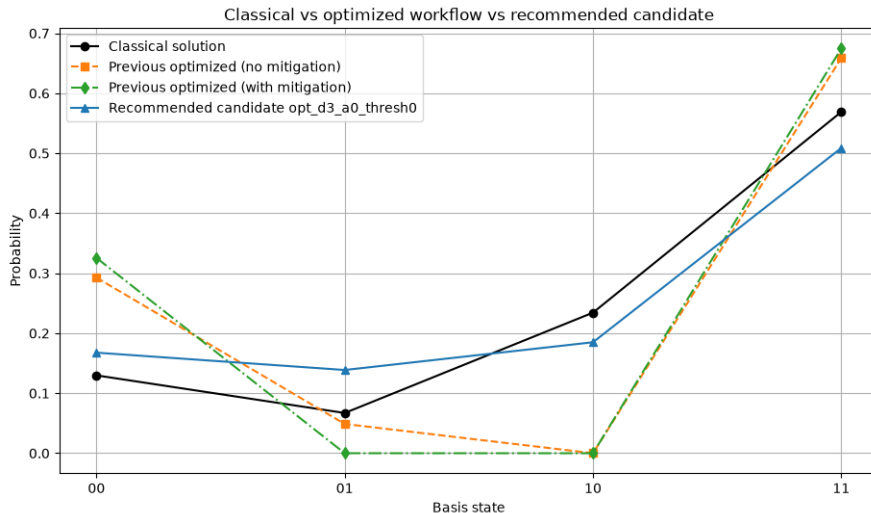
HHL algorithm steps



Typically, we need to convert a non-Hermitian matrix equation into a Hermitian one to apply HHL. This can be done by constructing an augmented matrix:

$$\tilde{A} = \begin{pmatrix} O & A \\ A^\dagger & O \end{pmatrix}, \tilde{\mathbf{b}} = \begin{pmatrix} \mathbf{b} \\ \mathbf{0} \end{pmatrix}$$

HHL Results and Analysis



HHL algorithm preliminary results.

Why HHL is not practical for chemistry?

Chemistry is about accuracy and HHL is about speed. Overall HHL is not practical for chemistry because of the following reasons:

- HHL is not accurate enough for chemistry. It is a probabilistic algorithm and the accuracy depends on the number of measurements and the condition number of the matrix.
- HHL is not suitable for NISQ devices. It requires a large number of qubits and gates, which are not available on current quantum computers.
- Also, in some cases, high circuit depth is required, which is not feasible on NISQ devices.
- HHL is not suitable for large matrices. It requires a large number of qubits and gates, which are not available on current quantum computers.
- Also, condition number of the matrix is important for HHL.
-

Sample-based Quantum Diagonalization Algorithm for solving system of linear equations

Pauli decomposition of A :

Any Hermitian matrix decomposes *exactly* in the **Pauli basis**:

$$A = \sum_P c_P P, \quad c_P = \frac{1}{N} \text{Tr}(P^\dagger A), \quad P \in \{I, X, Y, Z\}^{\otimes n}$$

Dimension counting: n qubits $\Rightarrow N = 2^n$ -dim. space $\Rightarrow 4^n$ possible Pauli strings. For **sparse / structured** A , only $K \ll 4^n$ terms survive — K is what sets the circuit cost later, *not* 4^n .

Worked example (our 4×4 test matrix, $n = 2$):

$$A = \begin{pmatrix} 5 & 3 & 1 & -2 \\ 3 & 5 & 2 & -1 \\ 1 & 2 & 5 & -3 \\ -2 & -1 & -3 & 5 \end{pmatrix}$$

$$A = 5(I \otimes I) + 1(X \otimes Z) + 2(Y \otimes Y) + 3(Z \otimes X)$$

only **4** of **16** possible terms are non-zero — eigenvalues 1, 3, 5, 11.

SQD for $A\vec{x} = \vec{b}$: Encoding & Time Evolution

Encoding and Time Evolution:

Encode: normalize $|b\rangle = \vec{b}/\|\vec{b}\|$ and load onto $n = \log_2 N$ qubits — **state preparation**.

Time-evolution operator:

$$U(t) = e^{-iAt} = e^{-it \sum_k c_k P_k}$$

Pauli terms don't commute $\Rightarrow$ **Trotterize:**

$$e^{-iAt} \approx \left(\prod_{k=1}^K e^{-ic_k P_k t/S} \right)^S$$

Since $P^2 = I$: $e^{-i\theta P} = \cos \theta I - i \sin \theta P$ — each factor is **one native rotation gate**.

Gate count: $K \times S$ (polynomial) vs. $O(4^n)$ CNOTs for an arbitrary unitary.

SQD for $A\vec{x} = \vec{b}$:

Sampling to solution:

- **Sample:** measure $|\psi(t_j)\rangle = U(t_j)|b\rangle$ at M times $t_j \rightarrow$ bitstrings.
- **Aggregate:** pool bitstrings into a **configuration subspace** $\mathcal{C}$, size $R = |\mathcal{C}|$.
- **Classical solve:** restrict $A_{\text{sub}} = P^\dagger A P$, $b_{\text{sub}} = P^\dagger b$; **diagonalize & invert:**
 $x_{\text{sub}} = W\Lambda^{-1}W^\dagger b_{\text{sub}}$.
- **Lift back:** $\vec{x} \approx P x_{\text{sub}}$.

Advantage when $R \ll N$: subspace diagonalization stays cheap even though N is huge.



Numerical Example: Pauli Decomposition of 4×4 matrix A

$$A = \begin{pmatrix} 5 & 3 & 1 & -2 \\ 3 & 5 & 2 & -1 \\ 1 & 2 & 5 & -3 \\ -2 & -1 & -3 & 5 \end{pmatrix}$$

Size of A : 4×4 Hilbert space dim.: $N = 4$ Qubits: $n = 2$ Possible Pauli strings: $4^n = 16$

Non-zero coefficients $c_P = \frac{1}{N} \text{Tr}(P^\dagger A)$ (the other 12 of 16 vanish):

Pauli string P	Coefficient c_P
II	5.000000
XZ	1.000000
YY	2.000000
ZX	3.000000

$$A = 5(I \otimes I) + 1(X \otimes Z) + 2(Y \otimes Y) + 3(Z \otimes X)$$

Reconstruction check: $\|A - A_{\text{reconstructed}}\|_F = 0.000000 \checkmark$

Numerical Example: Eigendecomposition of A

The eigendecomposition $A = V\Lambda V^\dagger$ gives:

k	λ_k	$ \phi_k\rangle$
0	1.0	$(-0.5, +0.5, -0.5, -0.5)$
1	3.0	$(-0.5, +0.5, +0.5, +0.5)$
2	5.0	$(+0.5, +0.5, -0.5, +0.5)$
3	11.0	$(-0.5, -0.5, -0.5, +0.5)$

Key structural fact: every eigenvector has entries ± 0.5 — all eigenvectors have **equal magnitude** entries. This is why $\vec{b} = (0, 0, 0, 1)$ overlaps *equally* with every eigenvector ($|\beta_k|^2 = 0.25$ for all k), so the sampled subspace never truncates below N for this particular example.

State Preparation & Eigenbasis Expansion

$\vec{b} = (0, 0, 0, 1)$ is already normalized ($\|\vec{b}\| = 1$). Represent it as a **quantum state** in the computational basis:

$$|b\rangle = b_0 |00\rangle + b_1 |01\rangle + b_2 |10\rangle + b_3 |11\rangle$$

Initial state:

$$|\psi_0\rangle = \frac{|b\rangle}{\|\vec{b}\|} = (0, 0, 0, 1) = |11\rangle$$

Expand in the eigenbasis of A :

$$|\psi_0\rangle = \sum_{k=0}^3 \beta_k |\phi_k\rangle, \quad \beta_k = \langle \phi_k | \psi_0 \rangle$$

k	β_k	$ \beta_k ^2$	λ_k
0	-0.5000	0.2500	1.0000
1	+0.5000	0.2500	3.0000
2	+0.5000	0.2500	5.0000
3	+0.5000	0.2500	11.0000

Structural caveat: all four overlaps are **exactly equal** — $\vec{b}$ has **no preferred eigenvector direction**, so sampling cannot discover a small dominant subspace for this particular example.

Numerical Example: Trotterized Time Evolution

For each of $M = 8$ time samples t_j linearly spaced in $(0, 2.0]$, the circuit prepares:

$$|\psi(t_j)\rangle = e^{-iAt_j} |\psi_0\rangle = \sum_k \beta_k e^{-i\lambda_k t_j} |\phi_k\rangle$$

The **Trotterized approximation** with $S = 4$ steps replaces the exact exponential with:

$$e^{-iAt} \approx \left(e^{-i \cdot 5 \cdot (II) \cdot t/S} \cdot e^{-i \cdot 1 \cdot (XZ) \cdot t/S} \cdot e^{-i \cdot 2 \cdot (YY) \cdot t/S} \cdot e^{-i \cdot 3 \cdot (ZX) \cdot t/S} \right)^S$$

Each factor is a single **native Pauli rotation gate**, so each circuit uses $4 \times S = 16$ rotation gates total — far fewer than the exponential gate count an arbitrary `UnitaryGate` would require.

Initial state (from Stage 2): $|\psi_0\rangle = |11\rangle$, amplitude 1.0000 on $|11\rangle$, zero elsewhere — unchanged at $t = 0$ before any evolution is applied.

Numerical Example: Sampled Probabilities Across Time

The computational basis of two-qubit states is $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.

The **probability** of measuring each basis state at time t_j is:

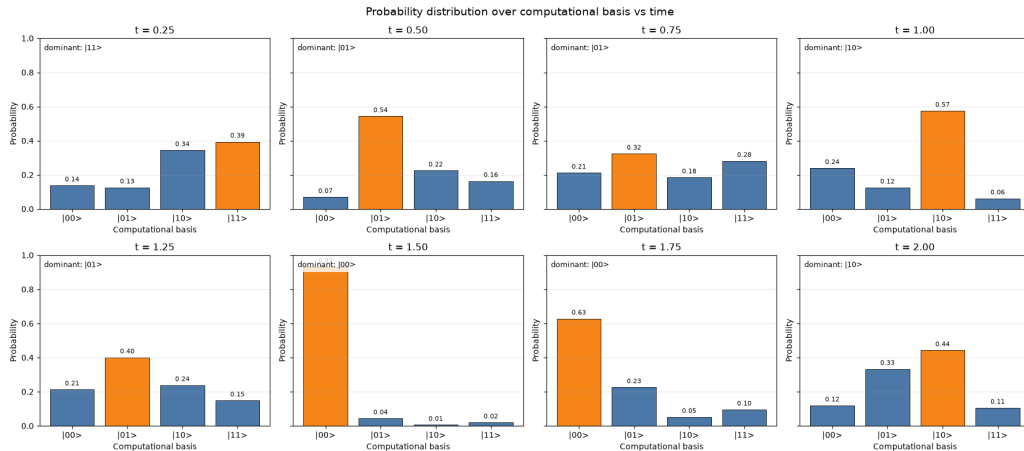
$$p(|00\rangle) = |\langle 00|\psi(t_j)\rangle|^2, \quad p(|01\rangle) = |\langle 01|\psi(t_j)\rangle|^2, \quad p(|10\rangle) = |\langle 10|\psi(t_j)\rangle|^2, \quad p(|11\rangle) = |\langle 11|\psi(t_j)\rangle|^2$$

Measuring $|\psi(t_j)\rangle$ at each of the 8 time samples gives:

t_j	$p(00\rangle)$	$p(01\rangle)$	$p(10\rangle)$	$p(11\rangle)$	Dominant
0.25	0.137	0.125	0.343	0.394	$ 11\rangle$
0.50	0.069	0.543	0.225	0.163	$ 01\rangle$
0.75	0.212	0.323	0.184	0.281	$ 01\rangle$
1.00	0.239	0.125	0.575	0.061	$ 10\rangle$
1.25	0.213	0.401	0.238	0.148	$ 01\rangle$
1.50	0.932	0.043	0.006	0.019	$00\rangle$ (sharp)
1.75	0.627	0.226	0.052	0.095	$ 00\rangle$
2.00	0.119	0.333	0.442	0.105	$ 10\rangle$

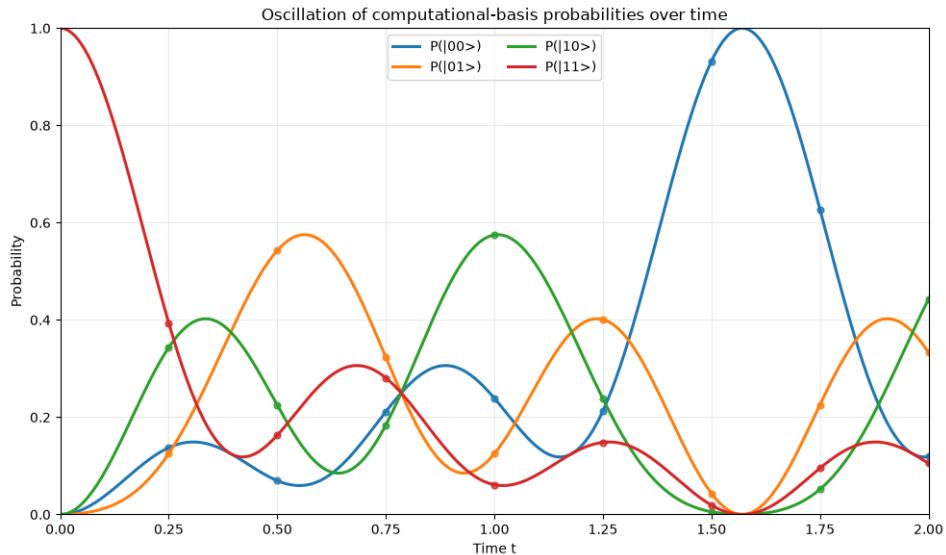
Observation: probabilities **oscillate strongly** with t_j — no single time reveals the full eigenstructure. At $t_j = 1.50$ the state concentrates **93.2%** on $|00\rangle$; this is exactly why sampling at **multiple times** is needed to discover all relevant configurations.

2D plots at each time moment for probability vs computational basis



Probability distribution across computational basis states at each time sample t_j .

Probability oscillations vs computational basis



Numerical Example: Aggregated Configuration Set

Pooling all **16,000 shots** across the 8 time samples:

Basis state	Total count	Empirical probability
$ 00\rangle$	5087	0.3179
$ 01\rangle$	4214	0.2634
$ 10\rangle$	4103	0.2564
$ 11\rangle$	2596	0.1623

All four basis states appear. The **configuration set** is:

$$\mathcal{C} = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\} \Rightarrow R = 4 = N$$

As predicted from the uniform $|\beta_k|^2 = 0.25$ overlap (Stage 2): for this particular $\vec{b}$, the sampled subspace **never truncates below the full space** — no compression, but a clean validation that the pipeline correctly recovers every relevant configuration.

Numerical Example: Subspace Projector & Restriction

Since $\mathcal{C}$ is the **full index set** $\{0, 1, 2, 3\}$, the projector P (never materialised as a dense matrix in code — built via sparse index slicing) selects *all* rows/columns of A :

$$P = P^\dagger = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = I_4$$

Restriction:

$$A_{\text{sub}} = P^\dagger A P = A, \quad b_{\text{sub}} = P^\dagger b = b$$

Verification output:

A_sub equals A: **True** b_sub equals b: **True**

The subspace restriction is a **no-op** here. This is the *correct* result for this toy matrix — the method is still valid, just not providing computational compression (since $R = N$, see Stage 5).

Numerical Example: Classical Diagonalization of A_{sub}

`np.linalg.eigh(A_sub)` returns:

$$\Lambda_{\text{sub}} = \text{diag}(1.0, 3.0, 5.0, 11.0)$$

$$W_{\text{sub}} = \begin{pmatrix} -0.5 & -0.5 & +0.5 & -0.5 \\ +0.5 & +0.5 & +0.5 & -0.5 \\ -0.5 & +0.5 & -0.5 & -0.5 \\ -0.5 & +0.5 & +0.5 & +0.5 \end{pmatrix}$$

These **match the true eigenvalues and eigenvectors of A exactly** — since $A_{\text{sub}} = A$ in this case (Stage 6).

Numerical Example: Spectral Inversion, Solve & Lift Back

$$x_{\text{sub}} = W_{\text{sub}} \Lambda_{\text{sub}}^{-1} W_{\text{sub}}^{\dagger} b_{\text{sub}}$$

Step 1 — project onto the eigenbasis (matches β_k from Stage 2 exactly):

$$W_{\text{sub}}^{\dagger} b_{\text{sub}} = \begin{pmatrix} -0.5 \\ +0.5 \\ +0.5 \\ +0.5 \end{pmatrix} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}$$

Step 2 — invert each eigenvalue, $1/\lambda_k$ (same inversion HHL performs *coherently* via controlled rotation + post-selection — SQD does it **classically**):

$$\Lambda_{\text{sub}}^{-1}(W_{\text{sub}}^{\dagger} b_{\text{sub}}) = \begin{pmatrix} -0.5/1.0 \\ +0.5/3.0 \\ +0.5/5.0 \\ +0.5/11.0 \end{pmatrix} = \begin{pmatrix} -0.5000 \\ +0.1\overline{6} \\ +0.1000 \\ +0.04\overline{5} \end{pmatrix}$$

Step 3 — reconstruct:

$$x_{\text{sub}} = W_{\text{sub}} \begin{pmatrix} -0.5000 \\ +0.1\overline{6} \\ +0.1000 \\ +0.04\overline{5} \end{pmatrix} = \begin{pmatrix} +0.1939 \\ -0.1394 \\ +0.2606 \\ +0.4061 \end{pmatrix} \xrightarrow{\text{lift back, } P=I} x_{\text{approx}} = x_{\text{sub}}$$

Matches full-space classical solution $x_{\text{classical}}$: True ✓

Numerical Example: Fidelity — Final Validation

Normalize both solution vectors:

$$\hat{x}_{\text{approx}} = \frac{x_{\text{approx}}}{\|x_{\text{approx}}\|}, \quad \hat{x}_{\text{classical}} = \frac{x_{\text{classical}}}{\|x_{\text{classical}}\|}$$

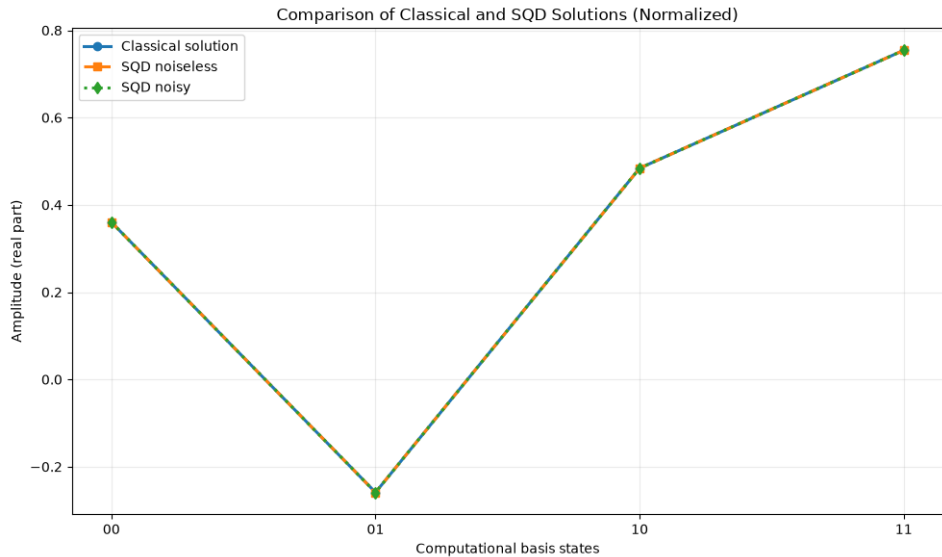
Compare via the **Bhattacharyya fidelity**:

$$F = \left(\sum_k \sqrt{|\hat{x}_{\text{approx}}|_k^2 \cdot |\hat{x}_{\text{classical}}|_k^2} \right)^2$$

$$F = 1.000000 \checkmark$$

The SQD solution matches the classical solution to machine precision.

Preliminary Results: SQD

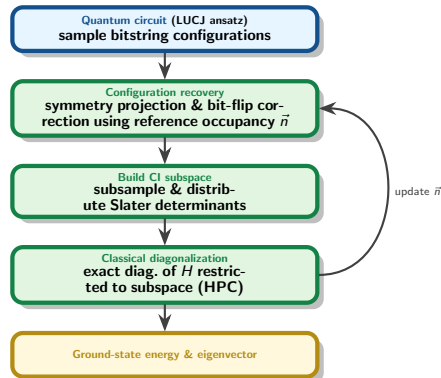


Why SQD is Better? SQD vs. HHL

Criterion	HHL	SQD
Complexity	$O(\log N \cdot \kappa^2 / \epsilon)$, but precision is paid for inside the circuit	$O(KS)$ quantum + $O(R^3)$ classical — precision moved to the cheap classical step
Transpilation	Arbitrary controlled- U $\Rightarrow O(4^n)$ CNOTs	Native Pauli rotations — Trotter gates transpile trivially
Gate count	QPE + QFT ⁻¹ + rotation + uncompute — deep	$K \times S = 16$ gates (our example) — no QPE/QFT machinery
Circuit depth	Scales with clock register — up to 11 qubits seen in practice	Fixed, shallow — no eigenvalue-precision qubits needed
Accuracy	Post-selection succeeds with $P \sim O(1/\kappa^2)$	$F = 1.000000$ achieved — exact classical diagonalization
Precision	Binary, set by clock width — more precision = deeper circuit	Exact, double-precision — classical, not qubit-limited

How does SQD work in chemistry?

- State prepared with a **LUCJ ansatz**, parameters set classically from CCSD t_2 -amplitudes — **no variational loop** needed.
- Sampled bitstrings are corrected via **configuration recovery**: symmetry projection (particle number, spin) and bit-flip correction using a reference occupancy vector $\vec{n}$.
- Recovered configurations build a CI subspace of Slater determinants; H is **diagonalized exactly** within that subspace on classical HPC nodes.
- $\vec{n}$ is updated from the result and the recovery loop repeats.
- Demonstrated on iron–sulfur clusters [2Fe–2S] (30e,20o) and [4Fe–4S] (54e,36o), and N_2 potential energy curves (cc-pVDZ) — up to **77 qubits** on a 133-qubit IBM Heron processor, median two-qubit gate fidelity **99.77%** [1].



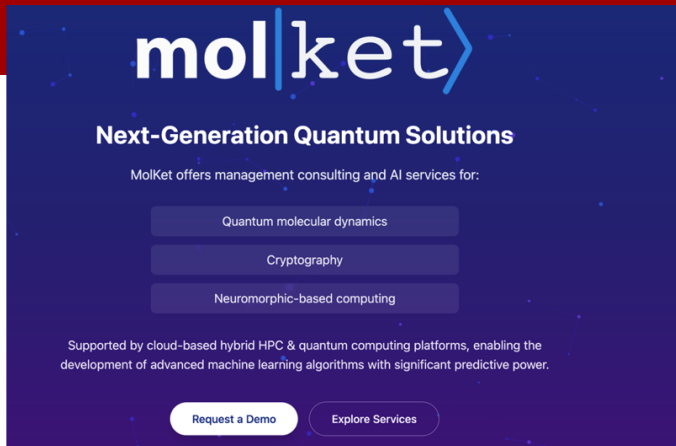
Current & Future Work

Current work:

- Running on IBM quantum computers and simulators to test the **SQD algorithm** for solving systems of linear equations.
- **Just received 180 minutes of approved QPU time** from IBM.
- **Long queue / waiting times** remain a practical bottleneck.

Future work:

- Test **scalability** and solution quality with increasing matrix size and qubit count.
- Test on **different quantum hardware** providers and architectures.
- **Benchmark** SQD against classical diagonalization and other quantum linear-system algorithms (HHL, VQLS).
- Analyze **computational complexity** and resource requirements for different matrix types and problem sizes.

The image shows a website banner for MolKet. At the top, the MolKet logo is displayed in white on a dark blue background. Below the logo, the text "Next-Generation Quantum Solutions" is written in white. Underneath, a line of text states "MolKet offers management consulting and AI services for:". This is followed by three stacked, rounded rectangular buttons with a dark blue background and white text. The buttons are labeled "Quantum molecular dynamics", "Cryptography", and "Neuromorphic-based computing". Below these buttons, a paragraph of text reads: "Supported by cloud-based hybrid HPC & quantum computing platforms, enabling the development of advanced machine learning algorithms with significant predictive power." At the bottom of the banner, there are two rounded rectangular buttons: a white one with dark blue text labeled "Request a Demo" and a dark blue one with white text labeled "Explore Services".

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Next-Generation Quantum Solutions

MolKet offers management consulting and AI services for:

- Quantum molecular dynamics
- Cryptography
- Neuromorphic-based computing

Supported by cloud-based hybrid HPC & quantum computing platforms, enabling the development of advanced machine learning algorithms with significant predictive power.

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The company also offers consulting services for training AI on datasets from fields such as chemistry, biology, materials science, and cosmology. These services are supported by physics-based kernels (e.g., quantum-mechanical kernels), enabling the development of advanced machine learning algorithms with significant predictive power.



Dr. Taha Selim

Directeur Général - General Manager

CREDENTIALS

- Lecturer/Researcher (QC Education Officer), Amsterdam University of Applied Sciences
- MVP Founder Member QSECDEF
- PhD Quantum Theoretical & Computational Chemistry
- M.Sc. of Quantum physics, Lasers, & Materials

RECENT PUBLICATIONS

- Carbon dioxide (CO₂) as a quantum molecular sensor in protoplanetary disks, at APS March Meeting 2024
- State-to-state rovibrational transition rates for CO₂ in the bend mode in collisions with He atoms, J. Chem. Phys., 159, 164310 (2023)

RECENT TALKS

- Co-speaker lightning talk Simulating quantum molecular dynamics with Julia at JuliaCon 2024
- Speaker, Carbon dioxide (CO₂) as a quantum molecular sensor in protoplanetary disks, at APS March Meeting 2024
- Panelist at the QED-C Quantum & AI Use Cases Panel (2025)



Alain Chancé

Président - President

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- Principal consultant with 30+ years experience in major management consulting firms
- MVP Founder Member QSECDEF
- IEEE Senior Member
- Society affiliate American Chemical Society (ACS)
- Member American Physical Society (APS)
- Qiskit Advocate & IBM Certified Associate Developer
- Master's Degree in Science and Executive Engineering MINES Saint-Etienne, France (1981)

RECENT PUBLICATIONS

- QPP-Alain, repository of the conference paper Quantum Permutation Pad with Qiskit Runtime
- Quantum Permutation Pad Corinthians, In: Femmam, S., Lorenz, P. (eds) Recent Advances in Communication Networks and Embedded Systems
- Quantum Chemistry and Computing for the Curious: Illustrated with Python and Qiskit® code, Keeper L. Sharkey, Alain Chancé, Packt (2022)

RECENT TALKS

- Co-speaker lightning talk Simulating quantum

Example projects

- Develop quantum algorithms and software for chemistry- and physics-inspired use cases across various quantum platforms, and we provide benchmarking services
- Train Smart AI-assistants for chemistry and physics simulations and design
- Automate Hamiltonian constructions with AI



contact:

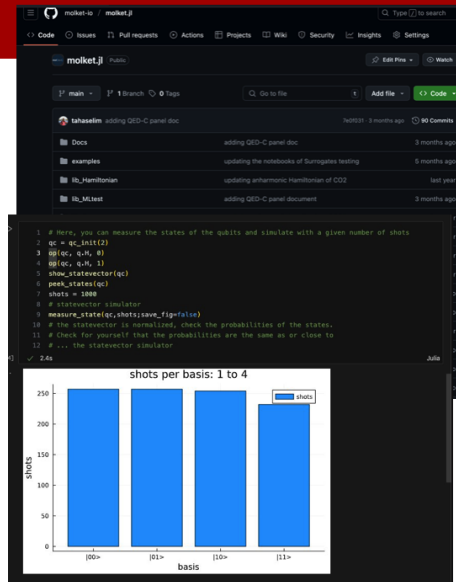
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- iQafé YouTube Channel:
<https://www.youtube.com/@iQafe>



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www.tahaselim.com

Thank you!

References

-  J. ROBLEDO-MORENO, M. MOTTA, H. HAAS, A. JAVADI-ABHARI, P. JURCEVIC, W. KIRBY, S. MARTIEL, K. SHARMA, S. SHARMA, T. SHIRAKAWA, I. SITDIKOV, R.-Y. SUN, K. J. SUNG, M. TAKITA, M. C. TRAN, S. YUNOKI, and A. MEZZACAPO, *Science Advances* **11** (2025).
-  X. COLLABORATION, The XRISM/Resolve view of the Fe K region of Cyg X-3, 2024.
-  J. DE PLAA, J. S. KAASTRA, L. GU, J. MAO, and T. RAASSEN, SPEX: High-Resolution Spectral Modeling and Fitting for X-ray Astronomy, 2019.
-  QULACS DOJO, 7.2 Harrow-Hassidim-Lloyd algorithm, https://dojo.qulacs.org/en/latest/notebooks/7.2_Harrow-Hassidim-Lloyd_algorithm.html, n.d., Accessed: 2025-11-23.